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Experimental and Theoretical Study of Tunable 1,3-Lithium Shift of Propargylic/Allenylic Species, Transmetallation, and Pd-Catalyzed Cross-Coupling Reactions

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Abstract: The highly selective tuning of the isomerization from 1-arylalka-1,2-dien-1-yllithium to 1-arylalka-1,2-dien-3-yllithium has been realized in the deprotonation of 1-arylalk-1-yne (conditions A and B) and carbolithiation of 1-arylbut-3-en-1-yne with alkyllithium (conditions C and D). Subsequent transmetallation and Pd-catalyzed Negishi coupling reactions afforded 1,1-diaryl or 1,3-diaryl allenes with high selectivity. Deuterium-labeling cross experiments indicated that an intermolecular lithiation process occurred in both 1,3-lithium shift conditions (conditions B and D). 1-Arylalka-1,2-diene was confirmed experimentally to be the intermediate. A computational study at the B3LYP level for the isomerization indicated that the acidity of H at the 3-position is higher than that of the H at the 1-position of 1-phenyl-1,2-butadiene. Under conditions B, iPr$_2$NH acts as a proton carrier to finish the 1,3-lithium shift. The overall activation barrier for the rate-determining step in the solvated models is $\approx$21.0 kcal mol^{-1}, indicating that the isomerization is reasonable at room temperature. For the isomerization under conditions D, DFT calculations indicated that the addition of TMEDA (tetramethylethylenediamine) and HMPA (hexamethylphosphoramide) changes the global minimum of the system; among the possible mechanisms (P1–P5) considered, the mechanism catalyzed by dilithiated species (P5) is the most probable one. The overall activation barriers for isomerization in THF and TMEDA solvated models are 22.6 and 19.7 kcal mol^{-1}, respectively, proving that the isomerization may proceed at RT in THF or at −78 °C with TMEDA, due to the fact that the solvation of the additives may increase the concentration of 1-phenyl-1,2-butadienyllithium monomer by a deaggregation effect.

Keywords: allenes · cross-coupling · density functional calculations · lithiation · reaction mechanisms

Introduction

Selectivity control in reactions of the same materials to form different products has attracted much attention in organic synthesis.[1] Palladium-catalyzed coupling reactions between an organic halide or triflate and an organometallic reagent have become one of the most commonly used C–C bond-forming reactions, in which there is usually no issue of regioselectivity involved.[2]

However, there will be such an issue if propargylic/allenylic species are applied, since both allenes and alkynes may be the products. In most cases, coupling reactions of propargylic/allenylic halides with Grignard reagents,[3] organozinc reagents,[4] organoboron reagents,[5] and organotin reagents[6] favor the formation of allene products, whereas alkynic products are formed if the steric hindrance of substrates or catalyst is appropriate.[7]

Recently, our group has reported that the propargylic/allenylic lithium formed by the lithiation of 1-aryl-1-alkyne would undergo sequential transmetallation and Pd-catalyzed coupling with organic halides leading to alkynes or allenes depending on the structures of the starting materials.[8] Furthermore, we also observed a ligand effect in the coupling reaction of propargyl/allenyl palladium species with organozinc reagents.[9] During our ongoing investigation, we observed the 1,3-lithium shift of 1-arylalka-1-yn-3-yl/1-arylalka-1,2-dien-1-yllithium forming 1-arylalka-1,2-dien-3-yl/1-

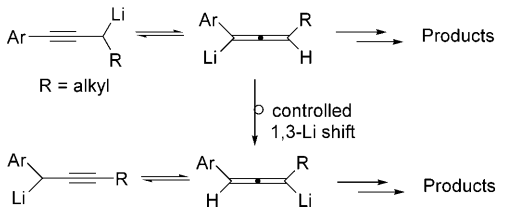
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arylalka-2-yn-1-yllithium (Scheme 1). This lithium shift can be controlled and followed by transmetallation and Pd-catalyzed coupling with halides to form 1,1-diaryl allenes or 1,3-diaryl allenes with high selectivity.[10]

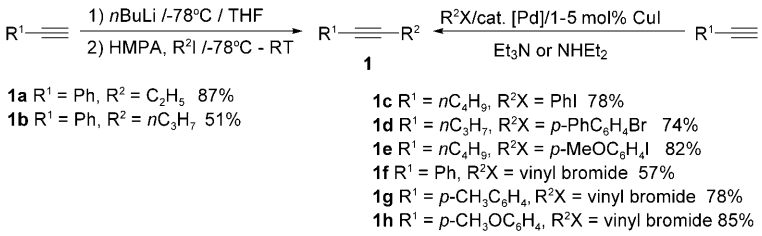


Scheme 1. The observed 1,3-Li shift of propargyl/1,2-allenyllithium species.

Although highly selective synthesis of allenes from propargylic/allenylic metallic derivatives has been widely studied,[11] only very few examples on 1,3-lithium-shift-based selective allene formation[12] have been reported. Most recently, E. Vrancken and co-workers reported their preliminary studies on the mechanism of this LDA-mediated 1,3-Li shift process of propargyl/1,2-allenyllithium species during the preparation of this manuscript:[13] they excluded the possibility of intramolecular and stepwise shift pathways based on DFT calculations (at the B3LYP/6-31+G** level), and proposed that LDA/iPr$_2$NH might act as a proton shuttle on the basis of experimental proof. This promoted us to present our own study on this reaction: the scope and mechanistic aspects of the 1,3-lithium shift and related transmetallation/cross-coupling reaction.

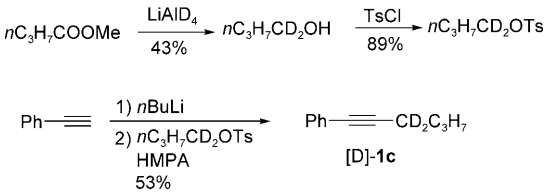
Results and Discussion

Preparation of the starting materials 1a–1h: The starting 1-aryl-1-alkynes **1a–h** were easily prepared by the reaction of arylethynyllithium with alkyl iodides[14] or the Sonogashira coupling of terminal alkynes with organic halides[15] (Scheme 2).



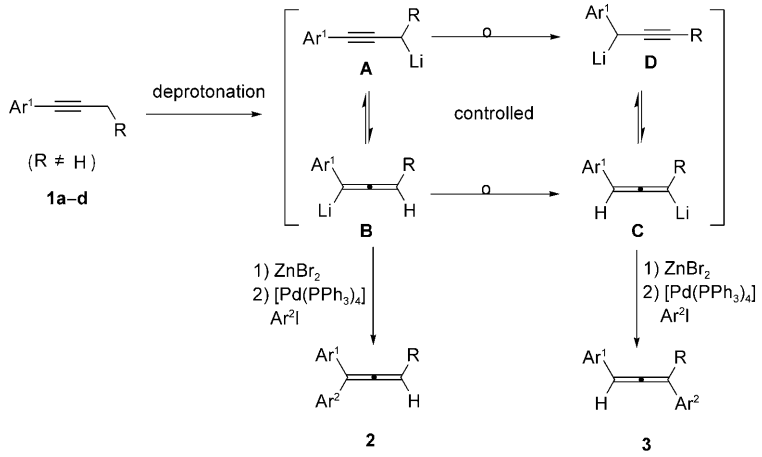
Scheme 2. Synthesis of starting materials **1a–h**.

The deuterated compound [D]-**1c** was synthesized by a three-step protocol in which the deuterium was introduced by the reaction of methyl butyrate with $LiAlD_4$ (Scheme 3).[16]



Scheme 3. Synthesis of [D]-**1c**.

1,3-Lithium shift of propargylic/allenylic lithium in lithiation of 1-aryl-1-alkynes: We have developed two sets of conditions to control the 1,3-lithium shift of propargylic lithiums **A** or 1,2-allenylic lithiums **B** formed by lithiation of 1-aryl-1-alkynes[10] (the alkyl group is beyond CH_3). By applying conditions A (lithiation at −20 °C with nBuLi for 1 h followed by transmetallation and coupling at RT), 1,1-diarylallenes **2** can be formed highly selectively. By conducting the reaction under conditions B (lithiation at RT with LDA for 1 h followed by transmetallation and coupling at RT), 1,3-diarylallenes **3** were produced exclusively (Scheme 4). The results summarized in Table 1 indicated that: 1) the reaction is general for differently substituted aryl halides and the regioselectivity for the formation of 1,1-diarylallenes **2** or 1,3-diarylallenes **3** is excellent; 2) the more sterically bulky aryl halides such as α-naphthyl iodide give the product in a lower yield (Table 1, entry 4); 3) both the aryl halides with an electron-donating group or an elec-

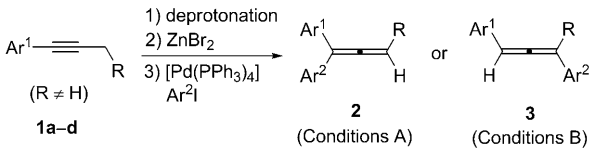


Scheme 4. A controlled 1,3-lithium shift reaction.

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Table 1. Sequential lithiation of 1-aryl-1-alkynes, transmetallation, and the subsequent Pd-catalyzed coupling reactions with aryl halides under conditions A and B.



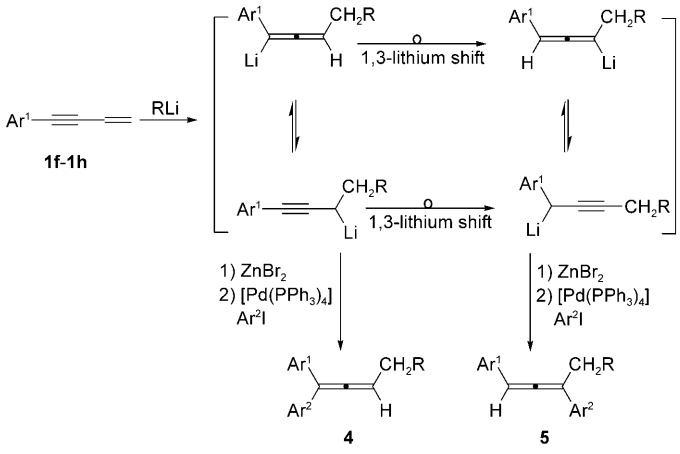
	1a–d Ar^1	CH$_2$R	Ar^2	Conditions A Yield of 2[a]	Conditions B Yield of 3[a]
1	Ph	C_2H_5 (**1a**)	Ph	55(**2a**)	84(**3a**)
2	Ph	nC_4H_9 (**1c**)	Ph	71[b](**2b**)	89(**3b**)
3	Ph	nC_4H_9 (**1c**)	*p*-MeC_6H_4	79[b](**2c**)	99(**3c**)
4	Ph	nC_4H_9 (**1c**)	α-naphthyl	63(**2d**)	56(**3d**)
5	Ph	nC_4H_9 (**1c**)	*p*-$MeOC_6H_4$	74[c](**2e**)	83(**3e**)
6	Ph	nC_4H_9 (**1c**)	*p*-$MeO_2CC_6H_4$	93(**2f**)	71(**3f**)
7	Ph	nC_3H_7 (**1b**)	Ph	76(**2g**)	96[d](**3g**)
8	*p*-PhC_6H_4	nC_3H_7 (**1d**)	Ph	77(**2h**)	72(**3h**)
9	*p*-PhC_6H_4	nC_3H_7 (**1d**)	*p*-BrC_6H_4	69(**2i**)	71(**3i**)

[a] Isolated yield. [b] The ratio of **2**/**3** = 99:1 as determined by 300 MHz ^{1}H NMR analysis. [c] The ratio of **2e**/**3e** = 98:2 as determined by 300 MHz ^{1}H NMR analysis. [d] NMR yield determined with CH_2Br_2 as the internal standard.

tron-withdrawing group could give the products in good to excellent yields (Table 1, entries 3, 5, and 6); 4) coupling with 4-bromophenyl iodide afforded the products **2i** and **3i** highly selectively, with the carbon–bromine bond untouched, in good yields (Table 1, entry 9).

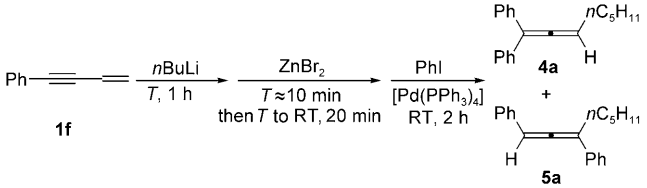
1,3-Lithium shift of propargylic/allenylic lithiums formed by conjugate addition of 1-arylbut-3-en-1-ynes with organolithiums: It has been reported that propargylic/allenylic lithium intermediates can also be formed by the conjugate addition of 1-arylbut-3-en-1-ynes **1f–h** with organolithium species.[17,18] Thus, we tried to extend this chemistry to the organolithiums formed by this conjugated addition reaction (Scheme 5).[19]

We initially tested the effect of temperature on the sequential reaction (Table 2). When 1-phenylbut-3-en-1-yne



Scheme 5. 1,3-Lithium shift of propargylic/1,2-allenylic lithiums formed by conjugate addition of 1-arylbut-3-en-1-ynes **1f–h** with organolithiums.

Table 2. The effect of temperature on carbolithiation of 1-phenylbut-3-en-1-yne with *n*BuLi, sequential transmetallation, and Pd^0-catalyzed cross-coupling with phenyl iodide.



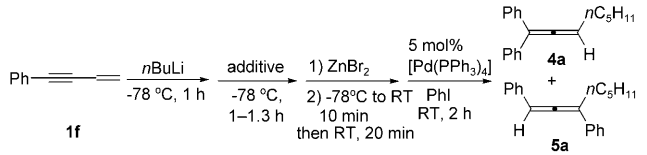
	T [°C]	**4a**/**5a**	Yield of **4a**+**5a** [%][a]
1	−78	100:0	54
2	−40	57:43	42
3	−20	14:86	44
4[b]	−78	0:100	18

[a] The yields are based on 1-phenylbut-3-en-1-yne and determined by 300 MHz ^{1}H NMR analysis with CH_2Br_2 as the internal standard. [b] After carbolithiation at −78 °C for 1 h, the reaction was warmed up to RT within a period of 2 h. Then, $ZnBr_2$ was added at room temperature, which was followed by Pd-catalyzed cross-coupling with phenyl iodide.

was treated with *n*BuLi at different temperatures, 1,1-diphenylocta-1,2-diene (**4a**) and 1,3-diphenylocta-1,2-diene (**5a**) were formed in different ratios. As shown in Table 2, a lower carbolithiation temperature favors the formation of **4a**, but a higher carbolithiation temperature favors the formation of the migrated product **5a**. When we raised the temperature to RT within 2 h after carbolithiation at −78 °C for 1 h, a complete 1,3-lithium shift occurred to afford **5a** as the only product, albeit in only 18% yield (Table 2, entry 4).

Some additives which may affect the 1,3-lithium shift reaction were screened: 1) pyridine, nBu_4NI, *N*,*N*-diisopropylethylamine, or DMAP (4-dimethylaminopyridine) have a limited effect on the 1,3-lithium shift (Table 3, entries 1–4); 2) although TMEDA (tetramethylethylenediamine) has

Table 3. The effect of additives on carbolithiation of 1-phenylbut-3-en-1-yne with *n*BuLi, and sequential transmetallation, Pd^0-catalyzed cross-coupling with phenyl iodide.



	Additive (1.2 equiv)	**4a**/**5a**[a]	Combined yield[b] [%]
1	pyridine	100:0	34
2	nBu_4NI	100:0	92
3	*N*,*N*-diisopropylethylamine	98:2	98
4	DMAP	95:5	98
5	TMEDA	0:100	≈50/≈70[c]
6	HMPA	100:0–0:100[d]	quantitative
7	HMPA[e]	0:100	83[f]

[a] The ratios of **4a** and **5a** are determined by 300 MHz ^{1}H NMR analysis. [b] The yields are based on aryl halides and determined by 300 MHz ^{1}H NMR analysis with CH_2Br_2 as the internal standard. [c] The isolated product was contaminated with an impurity. [d] The ratios changed constantly. [e] Two equivalents of HMPA were used (conditions D). [f] Isolated yield.

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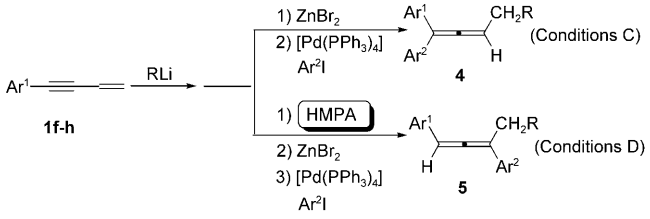
been proven to effect a clean 1,3-Li shift, the reaction was not easily reproducible with respect to chemical yields (Table 3, entry 5); 3) HMPA (hexamethylphosphoramide) has a similar effect to TMEDA on the 1,3-lithium shift (Table 3, entry 6). However, these reactions were surprisingly not easily reproducible even by the same chemist referring to the ratio of **4a**/**5a**. However, if the amount of HMPA was increased to two equivalents, reproducible results were obtained.

Through trial and error, we established a set of reproducible conditions to realize a clean 1,3-lithium shift in this protocol: after 1-phenylbut-3-en-1-yne was carbolithiated with *n*BuLi at −78 °C for 1 h, HMPA (2 equiv) was added at −78 °C, which was followed by warming up naturally to room temperature; subsequent transmetallation and Pd^0-catalyzed cross-coupling afforded **5a** exclusively, in 83 % yield (Table 3, entry 7), indicating a complete 1,3-lithium shift.

Thus, different conditions were established for the highly selective formation of 1,1-diarylallenes **4** or 1,3-diarylallenes **5** from the carbolithiation of **1f–h**. Conditions C afforded 1,1-diarylallenes **4**: 1-phenylbut-3-en-1-yne was carbolithiated with *n*BuLi at −78 °C for 1 h followed by transmetallation and coupling at RT. Conditions D afforded **5**: 1-phenylbut-3-en-1-yne was carbolithiated with *n*BuLi at −78 °C for 1 h, followed by addition of HMPA (2 equiv) at −78 °C; the solution was then warmed up naturally to room temperature, followed by transmetallation and coupling at room temperature.

Some typical results are summarized in Table 4, which indicate that: 1) primary, secondary and tertiary organolithium reagents may all be used; 2) the reaction is general for differently substituted 1-arylbut-3-en-1-ynes and aryl halides; 3) the switch is neat; and 4) the reactions all give good to excellent yields.

Table 4. Carbolithiation of 1-arylbut-3-en-1-yne with organolithiums, and sequential transmetallation-Pd^0-catalyzed cross-coupling with aryl halides under conditions C and D.



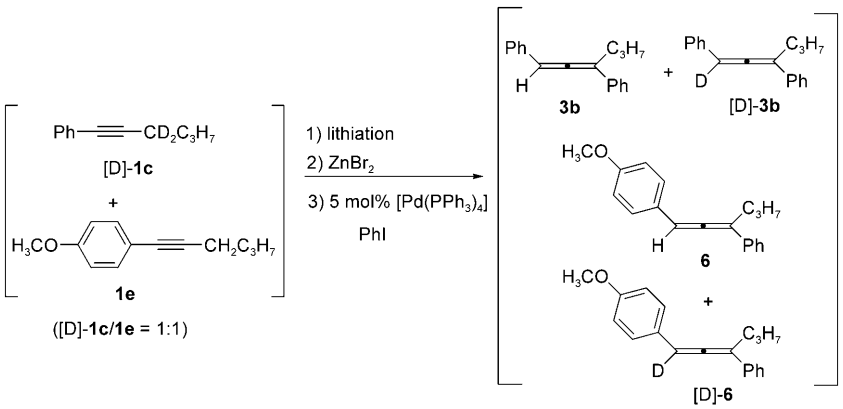
	1f–h Ar[1]	RLi R	Ar^2I Ar^2	Conditions C Yield of **4**[a]	Conditions D Yield of **5**[a]
1	Ph (**1f**)	*n*Bu	Ph	81 (**4a**)	83 (**5a**)
2	Ph (**1f**)	*n*Bu	*o*-$CH_3C_6H_4$	78 (**4b**)	75 (**5b**)
3	Ph (**1f**)	*n*Bu	α-naphthyl	92 (**4c**)	91 (**5c**)
4	Ph (**1f**)	*n*Bu	*p*-$MeOC_6H_4$	97 (**4d**)	98[b] (**5d**)
5	Ph (**1f**)	*n*Bu	*p*-$MeO_2CC_6H_4$	96 (**4e**)	93 (**5e**)
6	Ph (**1f**)	*n*Bu	*p*-CNC_6H_4	95 (**4f**)	95 (**5f**)
7	Ph (**1f**)	*s*Bu	Ph	84 (**4g**)	76 (**5g**)
8	Ph (**1f**)	*t*Bu	Ph	92 (**4h**)	85 (**5h**)
9	Ph (**1f**)	*t*Bu	*o*-$CH_3C_6H_4$	95 (**4i**)	79 (**5i**)
10	Ph (**1f**)	*t*Bu	α-naphthyl	99 (**4j**)	62 (**5j**)
11	Ph (**1f**)	*t*Bu	*p*-$MeOC_6H_4$	97 (**4k**)	69[c] (**5k**)
12	Ph (**1f**)	*t*Bu	*p*-NCC_6H_4	76 (**4l**)	69[c] (**5l**)
13	*p*-MeC_6H_4 (**1g**)	*n*Bu	Ph	89 (**4m**)	59 (**5m**)
14	*p*-$MeOC_6H_4$ (**1h**)	*n*Bu	Ph	61 (**4d**)	–

[a] Isolated yields. [b] The ratio **5d**/**4d** = 97:3. [c] Three equivalents of HMPA were used.

Mechanistic study of the 1,3-lithium shift: In order to study the real nature of the reaction, we first carried out a cross experiment for lithiation of a 1:1 mixture of [D]-**1c** and **1e** under conditions B. After sequential transmetallation and coupling with phenyl iodide catalyzed by Pd^0, a mixture of **3b**, [D]-**3b** , **6**, and [D]-**6** was produced. The lithiation by *n*BuLi and addition of HMPA also gave similar results, implying the possibility of an intermolecular mechanism (Scheme 6).

Based on these results, two pathways were proposed for the 1,3-lithium shift (Scheme 7): a) the iPr_2NH induced pathway; b) the bimolecular mechanism: intermolecular lithiation forming 1,3-dilithium intermediate **F** and 1-phenyl-1,2-diene **E**, followed by lithiation of **E** with **F** to finish the 1,3-shift. As described in detail in the computational parts in the following paragraphs, the iPr_2NH induced pathway was found to be in operation in the LDA-mediated isomerization process.

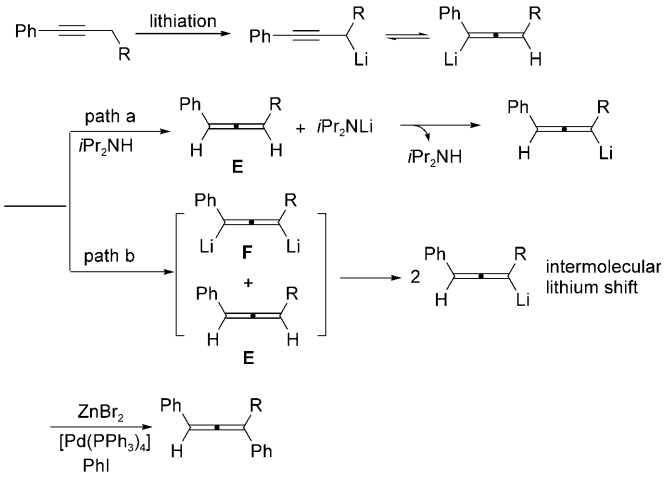
Thus, we designed the coupling reaction initiated by lithiation of 1-phenylocta-1,2-diene (**7**; an example of inter-



Scheme 6. Cross experiment reactions initiated with the lithiation of a 1:1 mixture of [D]-**1c** and **1e**. Lithiation conditions 1): LDA (2 equiv; prepared by mixing diisopropylamine and *n*BuLi with a molar ratio of 1:1.5); yield of **3b**+[D]-**3b**: 57 %; ratio **3b**/[D]-**3b**: 80:20; Yield of **6**+[D]-**6**: 78 %; ratio **6**/[D]-**6**: 71:29. The cross experiment was carried out under conditions B. Lithiation conditions 2): A mixture of [D]-**1c** and **1e** was lithiated with *n*BuLi at −20 °C for 1 h followed by addition of HMPA at −20 °C and warming up to room temperature over a period of 1 h. Subsequent transmetallation and Pd^0-catalyzed coupling reaction afforded the products. Yield of **3b**+[D]-**3b**: 36 %; ratio **3b**/[D]-**3b**: 58:42; Yield of **6**+[D]-**6**: 52 %; ratio **6**/[D]-**6**: 62:38.

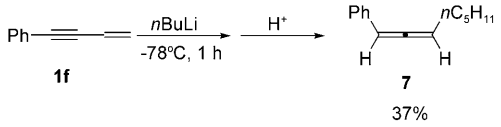
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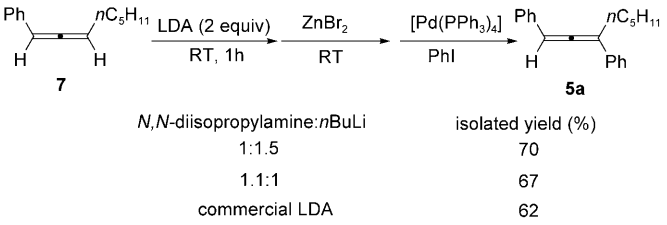
Scheme 7. Proposed pathways for the 1,3-lithium shift.

mediate allene **E**)—prepared by conjugate addition of 1-phenylbut-3-en-1-yne (**1f**) with *n*BuLi followed by protonation (Scheme 8)[16]—to study the possibility of path a. Com-



Scheme 8. Synthesis of 1-phenylocta-1,2-diene **7**.

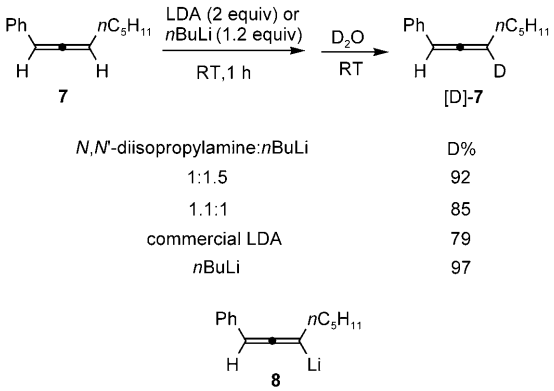
pound **7** was then lithiated by commercially available LDA (from Aldrich) or LDA prepared by mixing *N*,*N*-diisopropylamine with *n*BuLi for 1 h, followed by transmetallatation and Pd^0-catalyzed cross-coupling with phenyl iodide. As a result, 1,3-diphenylocta-1,2-diene **5a** was isolated in moderate yields (Scheme 9). These facts indicated that the 1,3-lithium shift may proceed via the **7**-type allene intermediate in



N,*N*-diisopropylamine:*n*BuLi	isolated yield (%)
1:1.5	70
1.1:1	67
commercial LDA	62

Scheme 9. Lithiation of 1-phenylocta-1,2-diene **7**, subsequent transmetallation, and Pd^0-catalyzed cross-coupling with phenyl iodide.

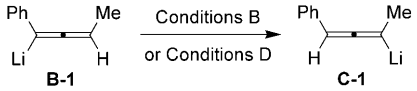
the presence of a base (Scheme 9). Furthermore, when D_2O was added after the treatment of 1-phenylocta-1,2-diene **7** with LDA, 1-phenyl-3-D-octa-1,2-diene [D]-**7** was formed, indicating the intermediacy of allenylic lithium **8** (Scheme 10).



N,*N*'-diisopropylamine:*n*BuLi	D%
1:1.5	92
1.1:1	85
commercial LDA	79
*n*BuLi	97

Scheme 10. Lithiation of 1-phenylocta-1,2-diene **7** and the subsequent quenching with D_2O.

Theoretical calculations: We used the simplest model intermediate **B-1** to study the mechanism of this 1,3-Li shift reaction by density functional (DFT) calculations with the Gaus-



Scheme 11. Isomerization of 1-phenyl-1,2-butadienyllithium.

sian 03 program[20] and the B3LYP[21] methods (Scheme 11). For each structure in the text, a harmonic vibration frequency calculation was conducted and thermal corrections were made. All structures were shown to be minima (having no imaginary frequencies) or first-order saddle points (transition states, having one imaginary frequency).

The mechanism under conditions B (deprotonation with LDA): In solution, there may be several solvated organolithium species: monomers, dimers, and higher oligomers. NMR studies indicate that LDA exits mostly in the form of dimers in THF.[22] However, monomers are usually more reactive than the dimer towards deprotonation,[23] which was also indicated in test calculations of these two pathways (see the Supporting Information). Therefore, in this study, the calculations of the deprotonation and isomerization reactions are all based on the monomeric organolithium model.

Gas-phase models: The reactants, intermediates, and transition states of deprotonation and isomerization were optimized at the B3LYP/6-31G** level[24a] and are shown in Figure 1.[25] Our test calculations show that the 6-31G** basis set is appropriate for this calculation (see the Supporting Information). To obtain more accurate energies, ΔG(6-311G**) was obtained by adding the thermal free energy corrections at the B3LYP/6-31G** level to the single point energy at the B3LYP/6-311G** level. The calculated relative free energies, ΔG(6-31G**) and ΔG(6-311G**), for each step of the deprotonation and isomerization process are

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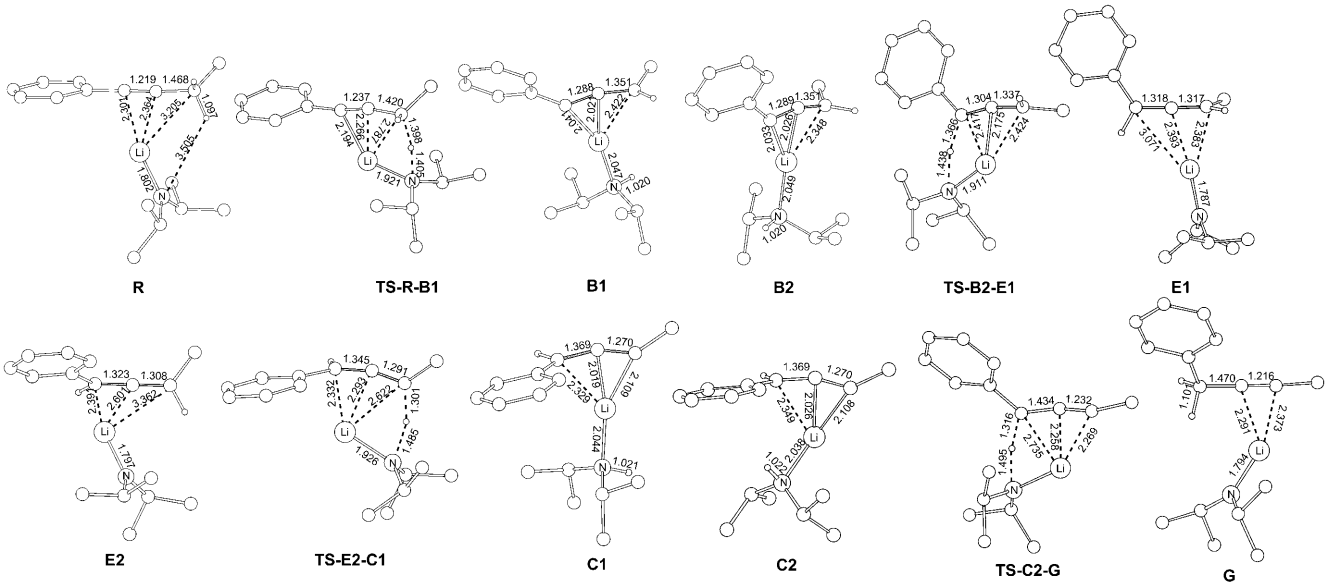


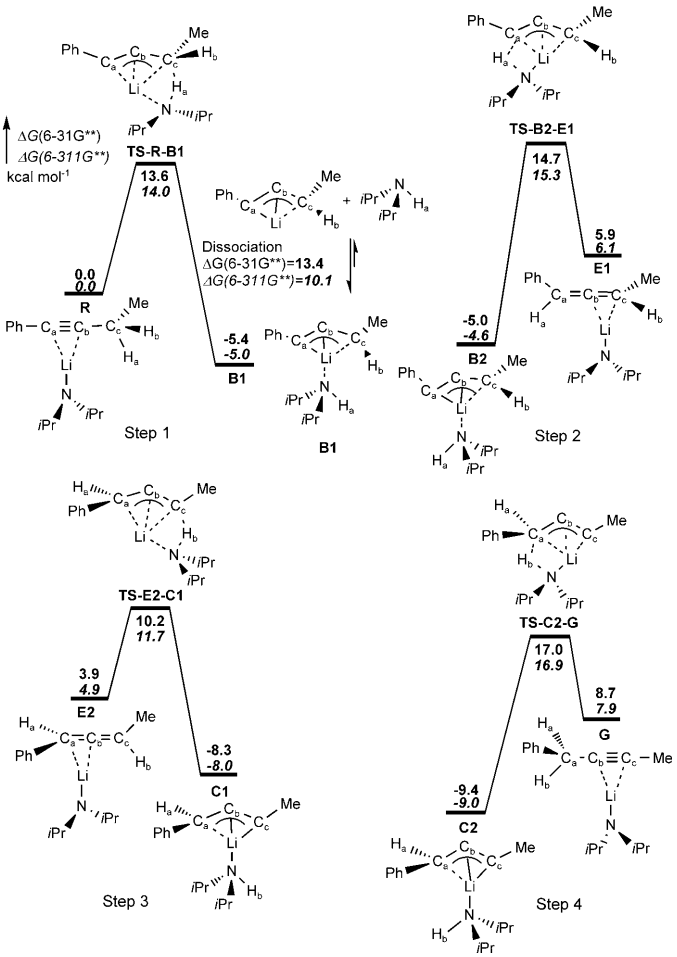
Figure 1. The optimized structures with selected bond lengths (Å) of the gas phase models shown in Scheme 12 (optimized at the B3LYP/6-31G** level). TS represents transition states. Hydrogen atoms not involved in the reactions are omitted for clarity.

shown in Scheme 12. **B1**/**B2** and **C1**/**C2** are conformational isomers of **B-1** and **C-1**, respectively, with the amine ligand.

LDA forms a complex with 1-phenyl-1-butyne, that is, **R**, and then the nitrogen atom of LDA abstracts H_a via transition state **TS-R-B1** yielding **B1**, which has a curved carbon skeleton.[26,27] Rotating the diisopropylamine moiety in **B1** 180° around the N–Li bond leads to conformer **B2**, in which H_a of the diisopropylamine is pointing to the left.

In step 2, C_a abstracts H_a connected to the nitrogen atom, producing the 1-phenylbuta-1,2-diene-LDA complex **E1** with an activation energy of 19.9 kcal mol^{-1}. By the end of this step, H_a has been transferred from C_c to C_a. Complex **E2** is a regioisomeric coordination complex of **E1**. In the third step N abstracts H_b from C_c, affording the isomerized product **C1** with a much lower activation energy (6.8 kcal mol^{-1}). Compound **C1** may isomerize to the more stable conformational isomer **C2**, which is 4.0 kcal mol^{-1} more stable than **B1**. A comparison between the activation energies of the three steps indicates that step 2 is the rate-determining step. The existence of the proposed transient intermediate 1-phenylbuta-1,2-diene and the validity of step 3 are supported by the experimental results presented in Schemes 9 and 10.

From our earlier reports it is known that, after being lithiated with BuLi or LDA, 1-phenyl-1-propyne can undergo a double 1, 3-hydrogen shift to form acetylide.[28] For **C2**, if the diisopropylamine molecule continues to transfer hydrogen to C_a, the final product will be an alkyne, that is, 1-phenyl-2-butyne. However, with a methyl here, the barrier for the second proton transfer (step 4 in Scheme 12) is higher (25.9 kcal mol^{-1}), and the alkyne product **G** is 16.9 kcal mol^{-1} less stable than **C2**. Therefore **C-1** is the final product of the isomerization. The corresponding activation barriers of step 2 in Scheme 12 with *t*BuLi and *n*BuLi are



Scheme 12. Reaction pathways and relative free energies ΔG(6-31G**) (in bold) and ΔG(6-311G**) (in italic) for gas phase models (in kcal mol^{-1}, 298 K), optimized at the B3LYP/6-31G** level.

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36.1 and 33.2 kcal mol^{-1},[24b] respectively, indicating that it is impossible for *t*BuLi and *n*BuLi to transfer protons as LDA does at RT.

The nature of the isomerization: It should be noted that the relative free energies of **TS-B2-E1** and **TS-E2-C1** are 15.3 and 11.7 kcal mol^{-1}, respectively (Scheme 12). A comparison of the energy of **B1** and **C2** (the most stable conformational isomers of **B-1** and **C-1**, respectively) suggests that **C-1** is 4.0 kcal mol^{-1} more stable than **B-1** in the gas phase, indicating a stronger acidity for H_b over H_a in **E1** and **E2**. Therefore, the nature of this isomerization is a series of acid-base equilibrium reactions.[29]

Solvated models: For the reaction in THF, the THF molecule and the Li ion have strong short-range effects, therefore the "mixed solvation model (explicit solvation model + continuum solvation model)" was used. For each structure the lithium atom was solvated with a THF molecule, and then the IEFPCM[30] (UAHF atomic radii) method was used to estimate the long-range effects in THF ($\varepsilon = 7.58$). Based on the gas-phase models, the solvated models for the intermediates and transition states included in steps 1–3 in Scheme 12 were constructed and optimized at the B3LYP/6-31G** level (Figure 2).[25] Then single-point energies and the long-range solvation effects were calculated at the B3LYP/6-311G** level. ΔG_{sol} was obtained by adding the thermal free energy corrections at the B3LYP/6-31G** level to the single-point energy and solvation corrections at the B3LYP/6-311G** level.

For the gas-phase models of **B1**/**B2** and **C1**/**C2**, an extensive conformational search was carried out to locate the linear allenyl or propargyl type structures, but the models all converged to bridged allenyl structures (Figure 1), which is consistent with previous calculations.[26] It should be noted that in the solvated models the relative free energies are within 2.4 kcal mol^{-1} for all the isomers of **B-1** and **C-1**. It is noted that minor changes in the relative position of the solvent molecules and the steric effect would lead to dramatic changes in the optimized structures of the intermediates, which are consistent with Reich's conclusion.[31] Compared with the gas phase models, the solvated models have some features (Figure 2): in addition to the bridged allenyl structure (*b*) (THF-**B1**-*b* and THF-**B2**-*b*), the linear allenyl (*a*) (THF-**B1**-*a*)- and propargyl (*p*) (THF-**B2**-*p*)-type isomers have also been located for **B-1** and THF-**B1**-*a* is the most stable; the *p*-type isomer is located for **C-1** (THF-**C1**-*p*) in addition to the *b* type isomer (THF-**C2**-*b*), which is the most stable.

A relaxed potential energy surface scan along the N–Li bond in **B1** in the gas phase shows that no transition state exists in the dissociation of **B1** to free *i*Pr$_2$NH and propargyl/1,2-allenyllithium species and the binding energy of the diisopropylamine to the lithium atom in **B1** is only 10.1 kcal mol^{-1} (step 1 in Scheme 12).[24b] For solvated models, the dissociation free energy of the coordinated diisopropylamine molecule from the lithium atom of THF-**B1**-*a* decreased further to 3.9 kcal mol^{-1} (at the B3LYP/6-31G** level), indicating that it is very easy for the diisopropylamine molecules to be exchanged intermolecularly. This accounts for the intermolecular proton transfer observed in the isotopic labeling cross experiments (Scheme 6).

As depicted in Figure 2, the three transition states of the solvated model, THF-**TS-R-B1**, THF-**TS-B2-E1**, and THF-**TS-E2-C1**, have the shortest Li–O bond lengths among all the structures listed in Figure 2. Thus, they are stabilized better than the other structures by THF. Consequently, compared with their gas phase counterparts, these three transition states have lower relative free energies (decreased by ≈4–7 kcal mol^{-1}). In conclusion, all the deprotonation and

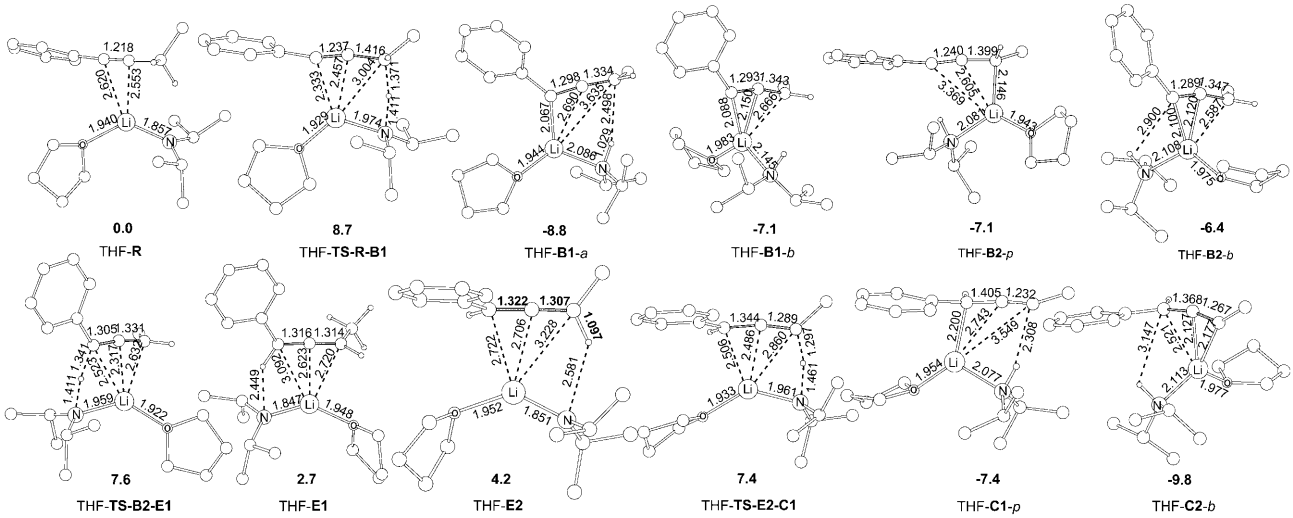


Figure 2. The optimized solvated species with selected bond lengths (Å) and relative free energies in solution ΔG_{sol} (in kcal mol^{-1}, 298 K). TS represents transition states. For allenyllithium intermediates: *a* represents linear allenyl structure, *b* represents bridged allenyl structure, *p* represents propargyl structure. Hydrogen atoms not involved in the reactions are omitted for clarity.

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isomerization steps (steps 1–3) in Scheme 12 are accelerated by the solvation of THF.

The activation energy for the rate-determining step (step 2, Scheme 12) is 16.4 kcal mol^{-1} in the solvated models (compare the energies of THF-**B1**-*a* and THF-**TS-B2-E1**, Figure 2). As an approximation, the overall activation energy could be viewed as the sum of the energies for dimer/monomer isomerization and the monomer-based activation barriers. After taking the energy difference between the dimer and monomer into account (approximately obtained by comparing the energy difference between THF2M*b* and THF2D in Figure 3), the overall activation barriers for the isomerization with LDA in solvated models is $\approx$21.0 kcal mol^{-1}, indicating that the isomerization is reasonable at room temperature in THF (Table 1).

The mechanism under conditions D (facilitated by the addition of TMEDA and HMPA): The models in this section were all optimized at the B3LYP/6-31G* level. ΔG(6-311G**) was obtained by adding the thermal free energy corrections at the B3LYP/6-31G* level to the single point energy at the B3LYP/6-311G** level.

The global minimum of the system: To understand the role of TMEDA and HMPA in the current reaction, identification of the global minimum of the system is necessary.[32] The alteration of the monomer/dimer equilibrium by additives may lead to a change in the global minimum of the system, which in turn could affect the overall activation barrier. Seven monomers (including bridged allenyl, linear allenyl, and propargyl structures) and three dimers with different numbers of explicit solvated THF molecules of 1-phenylbuta-1,2-dien-1-yllithium were used to find out the most stable species in THF (see the Supporting Information). The results show that in THF the most stable monomer (THF2M*b*, Figure 3) has a bridged allenyl structure, with the lithium atom being solvated by two THF molecules; the most stable dimer (THF2D, Figure 3) has a four-centered cyclic structure, solvated with a THF on each lithium atom, which is consistent with the NMR study.[22] The solvation numbers of the THF and HMPA dimers are all one solvent/lithium, as reported in the X-ray diffraction studies,[27,33] thus, the initial structures of Py (pyridine)-, TMEDA-, and HMPA-solvated monomers and dimers are constructed based on the THF-solvated structures. Two TMEDA-solvated dimers have been located: the bidentate model

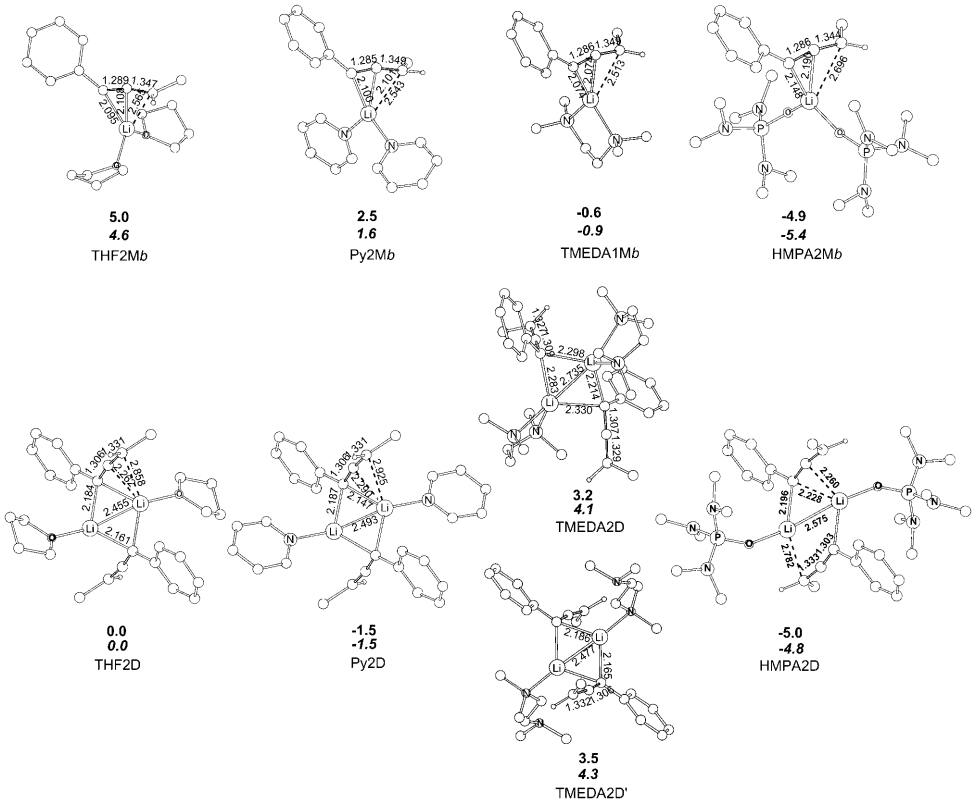


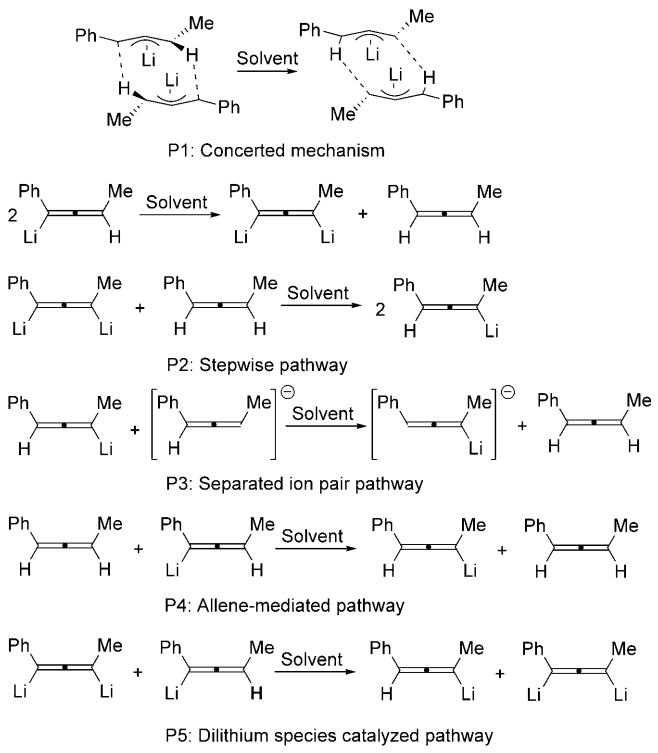
Figure 3. The optimized structures and relative free energies ΔG(6-31G*) (in bold) and ΔG(6-311G**) (in italic) (scaled to monomer, in kcal mol^{-1}, 298 K) of monomers (M) and dimers (D) solvated by THF, Py, TMEDA, and HMPA. These structures are named in this way (take THF solvated model for an example): THF2M*b* represents monomeric bridged allenyl structure solvated by two THF molecules; THF2D represents dimeric structure solvated by two THF molecules). The Py-, TMEDA-, and HMPA-solvated structures are named similarly. The energy of THF2D is set to zero energy. The relative energies are obtained by calculating the change of free energies of model reactions. Hydrogen atoms not involved in the reactions are not shown for clarity.

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(TMEDA2D) has a slightly lower free energy (0.2 kcal mol^{-1}) than the monodentate structure (TMEDA2D′, Figure 3). Generally, the solvation energies of monomers are higher than for dimers, but there is usually strong electrostatic stabilization in the latter.[34] The relative stabilities of monomers and dimers in solution depend on the structure of the carbanions and the solvents. The optimized dimer and monomer models with relative free energies ΔG(6-31G*) and ΔG(6-311G**) are given in Figure 3.[25] With THF and Py, dimeric 1-phenylbuta-1,2-dien-1-yllithium structures (THF2D and Py2D) are the global minima as their energies are lower relative to the corresponding monomeric structures, whereas with TMEDA and HMPA, monomeric structures (TMEDA1M*b* and HMPA2M*b*) are the global minima.

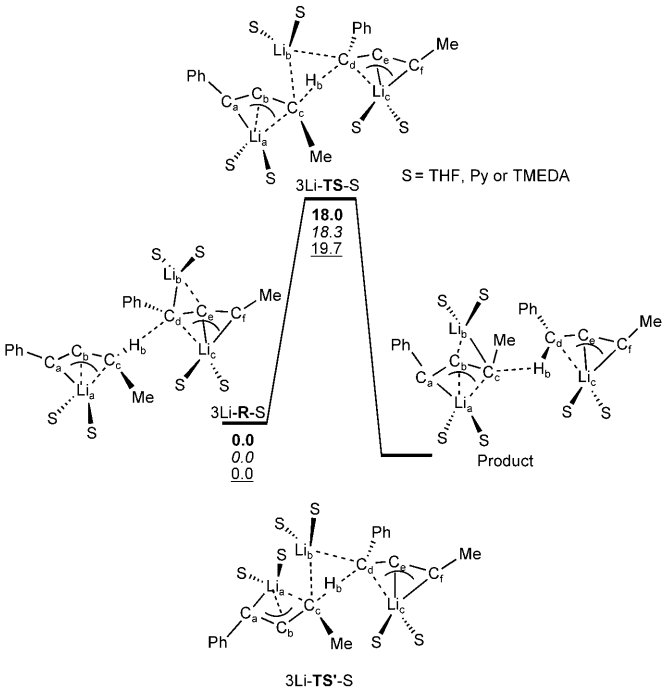
The catalytic dilithiated species mechanism: We again took the monomeric 1-phenylbuta-1,2-dien-1-yllithium as the reactive species in the calculation. There are several possible mechanisms for the isomerization under conditions D (Scheme 13): the calculation results show that the transition



Scheme 13. Possible mechanisms for the isomerization.

state for concerted double-hydrogen transfer[32] (P1) cannot be located; the overall activation energies for the first step of the stepwise pathway (P2) and the separated ion-pair-based mechanism[33] (P3) are over 30 kcal mol^{-1}, which is too high for an isomerization that proceeds at room temperature (entry 4, Table 2). In the presence of excessive BuLi under the experimental conditions, there should be higher concentrations of dilithium species instead of allene, therefore, the allene catalyzed pathway (P4) is not considered here (see the Supporting Information).

In the pathway P5, 1-phenylbuta-1,2-dien-1,3-yl dilithium may abstract the proton from 1-phenylbuta-1,2-dien-1-yllithium to finish the isomerization process and regenerate this dilithium intermediate. The optimized intermediates and transition structures of THF-, Py-, and TMEDA-solvated models for the pathway P5 shown in Scheme 14 are listed



Scheme 14. Representative sketch of the reaction pathway. The relative free energies ΔG(6-311G**) for THF- (bold), Py- (italic), and TMEDA- (underlined) solvated models are in kcal mol^{-1}.

in Figure 4.[25] HMPA models are too large and thus not calculated due to the limitation of the computer. In the transition state 3Li-**TS**-S (S = solvent), while the proton H_b transfers from C_c of the 1-phenylbuta-1,2-dien-1-yllithium moiety to C_d of the dilithiated species, Li_b shifts from the dilithiated species to the 1-phenylbuta-1,2-dien-1-yllithium; the structures for the transition state 3Li-**TS′**-S solvated by THF and TMEDA are less stable due to the repulsion between Li_a and Li_b (see the Supporting Information).

As shown in Figure 4, the energy differences between the three transition states of 3Li-**TS**-solvent, that is, 3Li-**TS**-THF, 3Li-**TS**-Py, and 3Li-**TS**-TMEDA, are within 1.7 kcal mol^{-1}. Similarly, if the energy difference between the dimer and monomer is taken into account (Figure 3), the overall activation barriers for the isomerization in THF- and TMEDA-solvated models are approximately 22.6 and 19.7 kcal mol^{-1}, respectively (Scheme 15), indicating that the isomerization is reasonable at room temperature in THF (entry 4, Table 2), and at a lower temperature (−78 °C,

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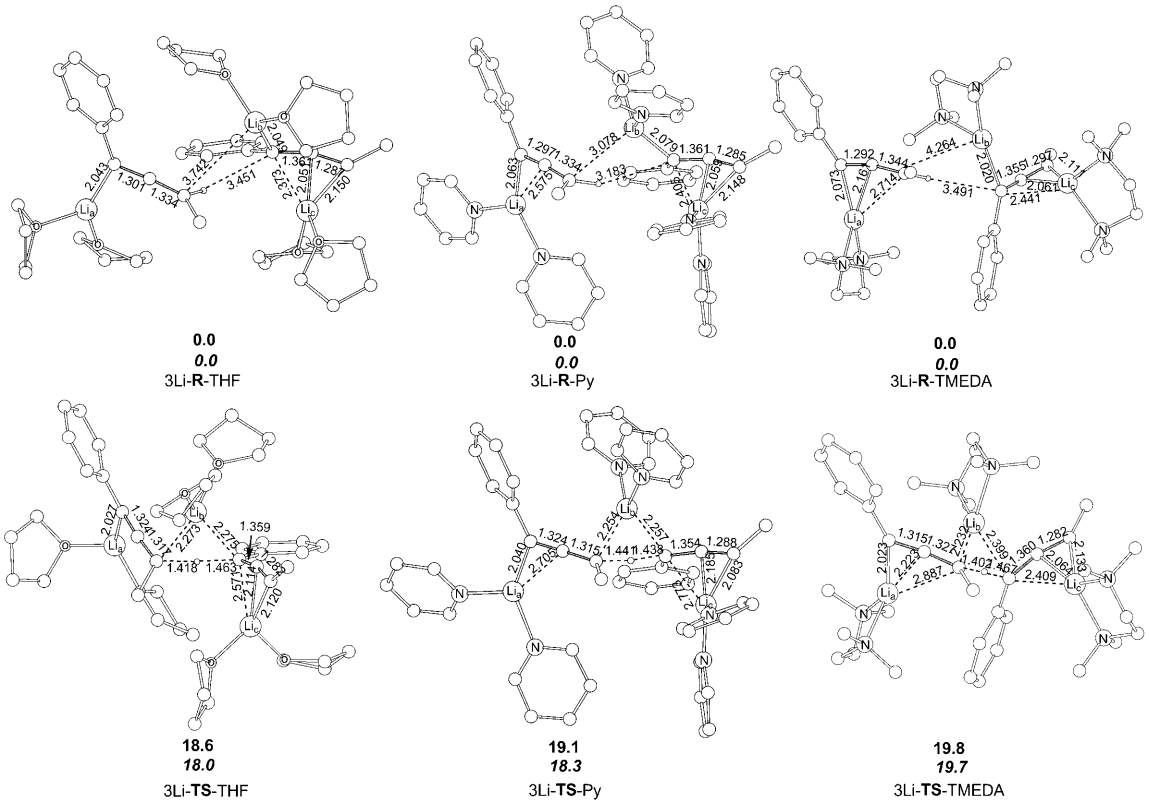
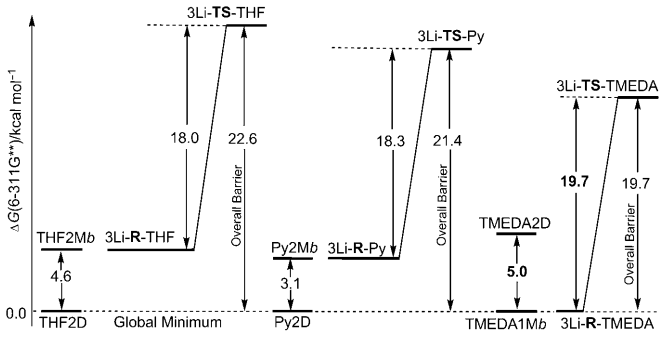


Figure 4. The optimized intermediates 3Li-**R**-solvent and transition states 3Li-**TS**-solvent (solvent = THF, Py, and TMEDA). The relative energies ΔG(6-31G*) (in bold) and ΔG(6-311G**) (in italic) are in kcal mol^{-1}. Hydrogen atoms not involved in the reactions are not shown for clarity.



Scheme 15. The overall activation barriers of THF-, Py-, and TMEDA-solvated models.

entry 5, Table 3) with TMEDA. The barrier for Py-solvated models (21.4 kcal mol^{-1}) is also consistent with the experimental observation that Py could not facilitate the isomerization at −78 °C (entry 1, Table 3). Therefore, it could be concluded that the isomerization under conditions D may be catalyzed by dilithiated species and accelerated by additives such as TMEDA and HMPA due to their deaggregation effect: with the stabilization effect of TMEDA (or HMPA), the monomeric species of 1-phenyl-1,2-butadienyl-1-lithium is dominant or has a remarkable concentration in the reaction solution. Consequently, the overall energy barrier for the H/Li shift reaction is reduced.

Conclusions

In conclusion, we have developed four protocols to control the 1,3-lithium shift. With the application of different reaction conditions, highly selective formation of 1,1-diaryl allenes or 1,3-diaryl allenes can be realized. Due to the easy availability and structural diversities of both starting compounds, this methodology will show its utility in organic synthesis. Experimental studies and DFT calculations suggested that LDA acts as a carrier to transfer the proton intra/intermolecularly with 1-phenylbuta-1,2-diene as the intermediate, and that the isomerization step is accelerated by THF solvation. Under conditions D, the reaction may be catalyzed by the 1-aryl-1,2-akadien-1,3-yl dilithium species. TMEDA and HMPA may increase the concentration of reactive monomers, accelerating the isomerization reaction through the deaggregation effect. Further studies in this area are being pursued in our laboratory.

Experimental Section

Syntheses of 1-(*p*-phenylphenyl)-1-(4′-bromophenyl)penta-1,2-diene (2i) and 1-(*p*-phenylphenyl)-3-(4′-bromophenyl)penta-1,2-diene (3i)

Conditions A: *n*BuLi (0.23 mL, 1.6 M in hexanes, 0.37 mmol) was added to a solution of 1-(*p*-phenylphenyl)pent-1-yne (68 mg, 0.31 mmol) in THF (3 mL) in a dry Schlenk tube at −20 °C under N_2. After being

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stirred at −20 °C for 1 h, a solution of dry $ZnBr_2$ (140 mg, 0.62 mmol) in THF (4 mL) was added at this temperature. After 5 min at this temperature the reaction mixture was warmed up to RT within 13 min, [Pd(PPh_3)$_4$] (18 mg, 5 mol %) and 4-bromoiodobenzene (105 mg, 0.37 mmol) were added subsequently at RT, and the resulting mixture was stirred at this temperature. After the reaction was complete, as monitored by TLC (eluent: petroleum ether (60–90 °C)), it was quenched with saturated NH_4Cl and extracted with diethyl ether. Drying over $MgSO_4$, filtration, rotary evaporation, and flash chromatography on silica gel (petroleum ether) afforded **2i** (80 mg, 69 %) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.65–7.20 (m, 13H), 5.79 (t, J = 6.6 Hz, 1H), 2.22 (quint, J = 7.5 Hz, 2H), 1.13 ppm (t, J = 7.5 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 205.1, 140.7, 140.0, 136.3, 135.8, 131.4, 130.0, 128.8, 128.6, 127.3, 127.1, 127.0, 120.9, 109.2, 96.6, 22.2, 13.5 ppm; MS (EI, 70 eV): *m/z* (%): 376 (96.17) [M^+(^{81}Br)], 374 (95.03) [M^+(^{79}Br)], 347 (95.18), 345 (100); IR (neat): $\tilde{\nu}$ = 1937, 1485 cm^{-1}; HRMS calcd for $C_{23}H_{19}Br$ [M^+]: 374.0670; found: 374.0654.

Conditions B: LDA (0.37 mL, 2.0 M in THF/ethyl benzene/pentane, commercial product from Aldrich, 0.73 mmol) was added to a solution of 1-(*p*-phenylphenyl)pent-1-yne (80 mg, 0.37 mmol) in THF (3 mL) in a dry Schlenk tube at −78 °C under N_2. The reaction was then warmed up to RT over a period of 1 h and then a solution of dry $ZnBr_2$ (407 mg, 1.81 mmol) in THF (4 mL) was added. After 25 min at this temperature, [Pd(PPh_3)$_4$] (21 mg, 5 mol %) and 4-bromoiodobenzene (207 mg, 0.73 mmol) were added, and the resulting mixture was stirred at RT. After the reaction was complete, as monitored by TLC, it was quenched with saturated NH_4Cl and extracted with ether. Drying over $MgSO_4$, filtration, rotary evaporation, and flash chromatography on silica gel (petroleum ether) afforded **3i** (97 mg, 71 %) as a solid. M.p. 107–108 °C (hexane); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.67–7.61 (m, 4H), 7.52–7.36 (m, 9H), 6.68 (t, J = 3.3 Hz, 1H), 2.72–2.51 (m, 2H), 1.28 ppm (t, J = 7.5 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 206.6, 140.8, 140.1, 135.2, 133.3, 131.6, 128.8, 127.7, 127.6, 127.3, 127.2, 127.0, 120.9, 111.1, 98.8, 23.2, 12.5 ppm; MS (EI, 70 eV): *m/z* (%): 376 (34.60) [M^+(^{81}Br)], 374 (34.46) [M^+(^{79}Br)], 45 (100); IR (neat): $\tilde{\nu}$ = 1930, 1488 cm^{-1}; elemental analysis calcd (%) for $C_{23}H_{19}Br$: C 73.61, H 5.10; found: C 73.58, H 5.15.

Synthesis of 1-phenyl-1-(2′-methylphenyl)octa-1,2-diene (4b) and 1-phenyl-3-(2′-methylphenyl)octa-1,2-diene (5b):

Conditions C: 1-Phenylbut-3-en-1-yne (52 mg, 0.41 mmol) in THF (4 mL) was added to a solution of *n*BuLi (0.30 mL, 1.6 M in hexanes, 0.49 mmol) in THF (2 mL) at −78 °C under N_2. After being stirred at −78 °C for 1 h, a solution of dry $ZnBr_2$ (186 mg, 0.82 mmol) in THF (4 mL) was added. After being stirred at −78 °C for 10 min, the reaction was allowed to warm up to RT over a period of 25 min; this was then followed by the addition of [Pd(PPh_3)$_4$] (16 mg, 5 mol %) and 2-methylphenyliodide (34 μL, 0.27 mmol) at RT with stirring. After the reaction was complete, as monitored by TLC, it was quenched with saturated NH_4Cl and extracted with diethyl ether. Drying over $MgSO_4$, filtration, rotary evaporation, and flash chromatography on silica gel (petroleum ether) afforded **4b** (58 mg, 78 %) as a liquid. ^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.35–7.13 (m, 9H), 5.61 (t, J = 6.9 Hz, 1H), 2.23–2.10 (m, 5H), 1.59–1.43 (m, 2H), 1.38–1.22 (m, 4H), 0.87 ppm (t, J = 6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$, 75.4 MHz): δ = 203.8, 137.3, 136.8, 136.5, 130.3, 130.2, 128.3, 127.4, 126.5, 126.4, 125.9, 107.6, 94.0, 31.4, 28.9, 28.8, 22.5, 20.1, 14.0 ppm; MS (EI, 70 eV): *m/z* (%): 276 (0.55) [M^+], 205 (100); IR (neat): $\tilde{\nu}$ = 1946, 1597, 1492, 1451 cm^{-1}; HRMS calcd for $C_{21}H_{24}$ [M^+]: 276.1878; found: 276.1907.

Conditions D: 1-Phenylbut-3-en-1-yne (52 mg, 0.41 mmol) in THF (4 mL) was added to a solution of *n*BuLi (0.31 mL, 1.6 M in hexanes, 0.49 mmol) in THF (2 mL) at −78 °C under N_2. After being stirred at −78 °C for 1 h, HMPA (0.14 mL, 0.82 mmol) was added and the reaction mixture was allowed to warm up naturally to RT over a period of 1 h; then a solution of dry $ZnBr_2$ (185 mg, 0.82 mmol) in THF (4 mL) was added. After 25 min, [Pd(PPh_3)$_4$] (16 mg, 5 mol %) and 2-methylphenyl iodide (34 μL, 0.27 mmol) were added at RT with stirring. After the reaction was complete, as monitored by TLC, it was quenched with saturated NH_4Cl and extracted with diethyl ether. Drying over $MgSO_4$, filtration rotary evaporation, and flash chromatography on silica gel (petroleum ether) afforded **5b** (56 mg, 75 %) as a liquid. ^{1}H NMR ($CDCl_3$, 300 MHz): δ = 7.39–7.23 (m, 5H), 7.23–7.16 (m, 4H), 6.25 (t, J = 3.0 Hz, 1H), 2.50–2.41 (m, 2H), 2.37 (s, 3H), 1.60–1.48 (m, 2H), 1.43–1.24 (m, 4H), 0.87 ppm (t, J = 7.1 Hz, 3H); ^{13}C NMR ($CDCl_3$, 75.4 MHz): δ = 203.9, 137.4, 135.9, 135.1, 130.5, 128.5, 128.0, 126.9, 126.8, 126.7, 125.8, 108.9, 95.1, 34.1, 31.6, 27.5, 22.5, 20.6, 14.1 ppm; MS (EI, 70 eV): *m/z* (%): 276 (5.70) [M^+], 205 (100); IR (neat): $\tilde{\nu}$ = 1943, 1598, 1456 cm^{-1}; HRMS calcd for $C_{21}H_{24}$ [M^+]: 276.1878; found: 276.1903.

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